

Introduction to Microeconomics –S102015CR

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Problem Set 2

Exercise 1

The cross elasticity of demand between cinema tickets and online streaming services is defined as

- a) the price elasticity of demand for cinema tickets divided by the price elasticity of demand for online streaming services.
- b) the change in the quantity of cinema tickets demanded divided by the change in the quantity of online streaming services demanded.
- c) the percentage change in the quantity of cinema tickets demanded divided by the percentage change in the price of online streaming services.
- d) the percentage change in the quantity of cinema tickets demanded divided by the percentage change in the quantity of online streaming services demanded.

Exercise 2

From Geneva, SWISS offers direct flights to Barcelona for CHF 500.- (round trip). During the last 3 months, an average of 10 passengers have purchased tickets for each day. For the Christmas holidays, SWISS has just launched a round trip offer for CHF 300.- thanks to which flights are full (20 passengers per flight) during the holidays. What is the price elasticity of demand (in absolute value)?

- a) 0.5
- b) 2
- c) 1
- d) 2.5

Exercise 3

Market research has shown that the demand function for soccer balls is as follows: $Q_D = 400 - 5P + 20P_B$, where P is the price of a soccer ball and P_B is the price of a basketball. The current price of a soccer ball is 20 and the price of a basketball is 15. At these price levels, what is the cross-price elasticity of demand for soccer balls with respect to the price of basketballs?

- a) 0.25
- b) 0.66
- c) 0.5

d) 0.33

Exercise 4 (January 2019 Exam)

Lilith spends all her income on comic books and ramen. Following the loss of her job, her income is cut in half and we see that her consumption of comics decreases and that of ramen increases. What can we deduce from this?

- a) Comics and ramen are both normal goods for Lilith.
- b) Comics are a normal good while ramen is an inferior good for Lilith.
- c) Comics are a luxury good for Lilith and ramen is a necessity good for Lilith.
- d) None of the above.

Exercise 5

Consider the demand for rental housing in San Francisco. The demand function is given by

$$Q_D = 1000 + 0.25R - P, \text{ with } R = 4000 \text{ and } P = 1600$$

1. Determine the quantity demanded.
2. Calculate the price elasticity of demand. Is demand elastic or inelastic?
3. Suppose rents increase by \$100. Does total spending increase or decrease?
4. Suppose the price is \$1000. Calculate the price elasticity of demand again. Is demand elastic or inelastic?
5. Suppose the price is \$800. Calculate the price elasticity of demand again. Is demand elastic or inelastic? And how does the total expense vary when the rent increases by \$100?

Exercise 6 (January 2017 Exam)

Consider the market for pencils characterized by the demand function $Q_D = 120 - 2P$ and the supply function $Q_S = 4P - 48$ where Q is the quantity and P is the market price. What are the price P , quantity Q , consumer surplus CS and producer surplus PS at market equilibrium?

- a) $P = 28, Q = 64, SC = 2048, SP = 512$.
- b) $P = 28, Q = 64, SC = 1024, SP = 512$.
- c) $P = 64, Q = 28, SC = 1024, SP = 512$.
- d) $P = 28, Q = 64, SC = 1025, SP = 1024$.

Exercise 7

Consider a market where, originally, the supply and demand functions are respectively: $Q_{S1} = 3P$ and $Q_D = 20 - P$. Following a technological change, the new supply function is: $Q_{S2} = 4P$. Which of the following statements is correct?

- a) Both before and after the technological change, the price elasticity of demand at equilibrium is equal to $-1/3$.
- b) Both before and after the technological change, the price elasticity of supply at equilibrium is equal to 1.
- c) After the technological change, the price elasticity of demand at equilibrium is equal to -4 .
- d) Before the technological change, the price elasticity of supply at equilibrium is equal to 3.

Exercise 8

Alsoft is a monopoly firm that provides highly specialized software. The unit price of its software is 75,000 francs. The willingness to pay (WTP) of potential customers is summarized by the table below:

Number of potential clients	WTP
1	120,000
1	100,000
1	75,000
1	50,000
1	25,000

Which of the following propositions is correct at the market equilibrium?

- a) The consumer surplus is 45,000 francs.
- b) The consumer surplus is zero.
- c) The consumer surplus will not change if the price increases to 75,001 francs.
- d) The consumer surplus is 70,000 francs.

Exercise 9 (January 2018 Exam)

The price elasticity of demand for good X is $\varepsilon_{p,X}^D = -0.5$, the price elasticity of demand for good Y is $\varepsilon_{p,Y}^D = -0.55$, and the cross-price elasticity of good X with respect to the price of good Y is negative ($\varepsilon_{p,Y}^{X} < 0$).

Which of the following options is correct?

- a) Goods X and Y are substitutes and the demand for good X is elastic.
- b) Goods X and Y are substitutes and the demand for good Y is inelastic.
- c) Goods X and Y are complements and the demand for good X is elastic.
- d) Goods X and Y are complements and the demand for good Y is inelastic.

Exercise 10

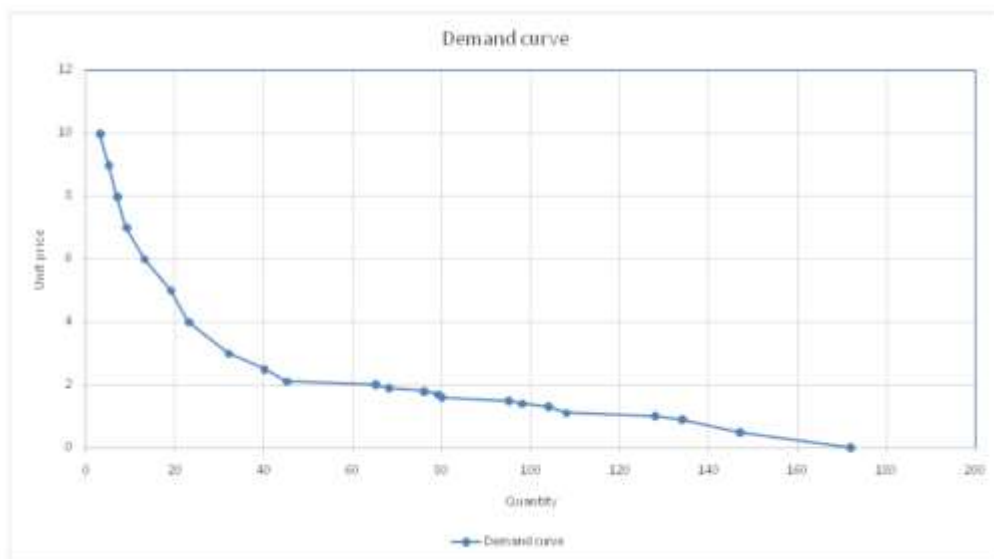
Among the following statements, which one is correct?

- a) The consumer surplus can be graphically shown by the area below the demand curve and above the firm's marginal cost.

- b) Consumer surplus equals producer surplus at equilibrium.
- c) The total surplus can be graphically shown by the area below the demand curve and above the firm's marginal cost.
- d) The producer surplus is equal to its p

Exercise 11

The figure below reports the demand curve of some good. At what combination of price and quantity does the seller maximize their revenue?



- a) At Price = 0 CHF (Quantity demanded is 172).
- b) At Price = 0.5 CHF (Quantity demanded is 147).
- c) At Price = 1.0 CHF (Quantity demanded is 128).
- d) At Price = 1.5 CHF (Quantity demanded is 95).
- e) At Price = 2.0 CHF (Quantity demanded is 65).